

Exercise sheet 1

1. Prove the following lower bound for the number of primes up to x :

$$\pi(x) \geq \log \log x.$$

Hint: Let p_n denote the n -th prime number. Try to use induction to show that

$$p_n \leq 2^{2^{n-1}}.$$

In particular, this implies that $p_n \leq e^{e^{n-1}}$. Then, try to take logarithm.

2. Prove the following identities:

(a)

$$\tau(n^2) = \sum_{d|n} \mu(d) \tau\left(\frac{n}{d}\right)^2$$

(b)

$$\sum_{d^2|n} \mu(d) = \mu(n)^2, \quad \sum_{d^k|n} \mu(d) = \begin{cases} 0, & \text{if } m^k|n, \text{ for some } m > 1, \\ 1, & \text{otherwise.} \end{cases}$$

(c)

$$\mu(n) = \sum_{\substack{1 \leq a \leq n \\ (a,n)=1}} e^{2\pi i \frac{a}{n}}$$

3. Prove that $\sum_{d|n} \mu(d)^2 = 2^{\omega(n)}$ and $\sum_{d|n} \mu(d) \tau(d) = (-1)^{\omega(n)}$.
4. Let f be a multiplicative arithmetic function. We already know that its Dirichlet inverse f^{-1} is multiplicative. Show that f^{-1} is completely multiplicative if and only if $f(p^\ell) = 0$ for all $\ell \geq 2$.
5. Show that there are exactly two arithmetic functions f such that $f * f = u$. Are they multiplicative?
6. Assume that f and g are such that $f(1) \neq 0$ and $g(1) \neq 0$. Prove that

$$(f * g)^{-1} = f^{-1} * g^{-1}.$$

7. For $k \geq 0$, define $\Lambda_k(n) = \sum_{d|n} \mu(d) \left(\log \frac{n}{d}\right)^k$. Show that $\Lambda_k(n)$ is supported on numbers n having at most k distinct prime factors, i.e.,

$$\Lambda_k(n) = 0, \quad \text{if } \omega(n) > k.$$

Moreover, prove that

$$0 \leq \Lambda_k(n) \leq (\log n)^k.$$

Exercise sheet 2

1. Find an arithmetic function f such that for all $n \geq 1$:

$$\frac{1}{\varphi(n)} = \sum_{d|n} \frac{1}{d} f\left(\frac{n}{d}\right).$$

2. Show that there exist constants c_1 and c_2 such that

$$\sum_{n \leq x} \frac{1}{\varphi(n)} = c_1 \log x + c_2 + O\left(\frac{\log x}{x}\right),$$

for all $x \geq 1$.

3. Show that for every $x \geq 3$, we have the asymptotic formula:

$$\sum_{2 < n \leq x} \frac{1}{n \log n} = \log \log x + C_1 + O\left(\frac{1}{x \log x}\right),$$

where $C_1 \in \mathbb{R}$ is a constant.

4. Let $r_2(n)$ be the arithmetic function which counts the number of representations of an integer $n \geq 1$ as the sum of two squares of integers

$$r_2(n) = \#\{(a, b) \in \mathbb{Z}^2, a^2 + b^2 = n\}.$$

Show that

$$\sum_{n \leq x} r_2(n) = \pi x + O(\sqrt{x}).$$

5. An integer is called square-free if it is not divisible by any square. Let

$$Q_2(x) = \#\{n \leq x : n \text{ is square-free}\}.$$

Show that there exists a constant C_2 such that

$$Q_2(x) = C_2 x + O(\sqrt{x}).$$

6. Show that for any $\varepsilon > 0$, there exists a positive constant C_ε such that

$$\tau(n) \leq C_\varepsilon n^\varepsilon$$

for all $n \geq 1$. Here $\tau(n)$ denotes the divisor function.

Exercise sheet 3

1. Let f be an arithmetic function satisfying

$$\sum_{p \leq x} f(p) \log p = (ax + b) \log x + cx + O(1), \quad \text{for } x \geq 2.$$

Show that there is a constant B (depending on f) such that for $x \geq 2$, one has

$$\sum_{p \leq x} f(p) = ax + (a + c) \left(\frac{x}{\log x} + \int_2^x \frac{1}{\log^2 t} dt \right) + b \log \log x + B + O\left(\frac{1}{\log x}\right).$$

2. Show that (for $n \geq 16$)

$$\sum_{p|n} \frac{1}{p} \ll \log \log \log n.$$

3. Show that the following series

$$\sum_p \frac{1}{p(\log \log p)^\alpha}$$

converges when $\alpha > 1$, and diverges when $\alpha \leq 1$.

4. Let $S(x)$ be defined by

$$S(x) := \#\{n \leq x : p|n \Rightarrow p \leq \sqrt{n}\}.$$

Prove the following asymptotic formula

$$S(x) = (1 - \log 2)x + O\left(\frac{x}{\log x}\right).$$

5. Show that as $x \rightarrow \infty$ the assertion

$$\sum_{n \leq x} \frac{\mu(n)}{n} = o(1)$$

would imply the prime number theorem.

6. Let $L(N) := [1, 2, \dots, N]$ the least common multiple of $1, 2, \dots$ and N . Show that the prime number theorem is equivalent to the following statement

$$\lim_{N \rightarrow \infty} L(N)^{\frac{1}{N}} = e,$$

where $e = 2.71\dots$ is the base of the natural logarithm.

Hint: perhaps there is a relation $L(N) \sim \sum_{n \leq N} \Lambda(n)$ in between?

7. Show that for $n > 1$, there is a prime p such that $p \parallel n!$ (recall that $p^r \parallel m$ denotes $p^r | m$ but $p^{r+1} \nmid m$.)

Exercise sheet 4

1. Let $f(n)$ be an arithmetic function and let $L_f(s)$ be its associated Dirichlet series. Show that the following two statements are equivalent:
- There exists $A > 0$ such that $f(n) \ll n^A$ for all $n \geq 1$.
 - The abscissa of absolute convergence of $L_f(s)$ is $< \infty$.

2. (a) Let $L_f(s)$ be the Dirichlet series associated to f . Find an arithmetic function $f(n)$ such that σ_c and σ_a satisfy the relation

$$\sigma_c = \sigma_a - 1.$$

Here σ_c (resp. σ_a) denotes the abscissa of *simple* (resp. *absolute*) convergence of $L_f(s)$.

(b) Let σ_a and σ_c be real numbers such that $\sigma_a - 1 \leq \sigma_c \leq \sigma_a$. Find an arithmetic function $f(n)$ such that $L_f(s)$ has the abscissa of absolute convergence σ_a and the abscissa of simple convergence σ_c .

3. Find the abscissa of absolute convergence for the following Dirichlet series

$$F(s) := \sum_{m_1, m_2, m_3 \geq 1} \frac{1}{(m_1 + m_2 + m_3)^s}.$$

Can you express $F(s)$ in terms of the Riemann zeta function $\zeta(s)$?

4. Let $\omega(n)$ and $\Omega(n)$ denote the number of distinct and total prime factors of n , respectively. If $n = p_1^{a_1} \dots p_k^{a_k}$ then $\kappa(n) = a_1 a_2 \dots a_k$ and $\kappa(1) = 1$. Express the following Dirichlet series in terms of the Riemann zeta function $\zeta(s)$.

$$\sum_{n=1}^{\infty} \frac{\tau(n^2)}{n^s}, \quad \sum_{n=1}^{\infty} \frac{2^{\omega(n)}}{n^s}, \quad \sum_{n=1}^{\infty} \frac{(-1)^{\Omega(n)}}{n^s}, \quad \sum_{n=1}^{\infty} \frac{3^{\omega(n)} \kappa(n)}{n^s}.$$

5. Recall for $c > 0$ we write $\int_{(c)} = \int_{c-i\infty}^{c+i\infty}$ for the contour integral oriented positively along the vertical line in $\mathbb{C}$ with real part equal to c .

(a) Show that

$$\frac{1}{2\pi i} \int_{(2)} \frac{y^s}{s^2} ds = \begin{cases} 0, & \text{if } 0 < y \leq 1, \\ \log y, & \text{if } 1 \leq y. \end{cases}$$

(b) Express

$$\frac{1}{2\pi i} \int_{(2)} \frac{y^s}{s^2} \left(-\frac{\zeta'}{\zeta}(s) \right) ds$$

as a finite sum involving the von Mangoldt function Λ .

6. Derive the following identities, valid for $\operatorname{Re}(s) > 1$.

$$\frac{1}{\zeta(s)} = s \int_1^{\infty} \frac{M(x)}{x^{s+1}} dx, \quad \text{where } M(x) = \sum_{n \leq x} \mu(n);$$

$$-\frac{\zeta'(s)}{\zeta(s)} = s \int_1^\infty \frac{\psi(x)}{x^{s+1}} ds, \quad \text{where} \quad \psi(x) = \sum_{n \leq x} \Lambda(n).$$

Exercise sheet 5

Definition. Let $k = 2^\alpha$ ($\alpha \geq 3$). For every odd a , there exists a pair $(r_{-1}(a), r_0(a))$ with $0 \leq r_{-1}(a) < 2$, $0 \leq r_0(a) < 2^{\alpha-2}$ such that

$$a \equiv (-1)^{r_{-1}(a)} 5^{r_0(a)} \pmod{2^\alpha},$$

where $(r_{-1}(a), r_0(a))$ is called the system of indices of a with respect to the modulus 2^α .

1. Assume that $\alpha \geq 2$. Prove the following

a) $r_{-1}(a) \equiv \frac{a-1}{2} \pmod{2}$.

b) Given a pair (r'_{-1}, r'_0) , the following two statements

$$(-1)^{r'_{-1}} 5^{r'_0} \equiv a \pmod{2^\alpha}$$

and

$$r'_{-1} \equiv r_{-1}(a) \pmod{2}, r'_0 \equiv r_0(a) \pmod{2^{\alpha-2}}$$

are equivalent.

Definition. Let k and k^* be positive integers such that $k^*|k$, and let $\chi \pmod{k}$ and $\chi^* \pmod{k^*}$ be Dirichlet characters. We say that χ is *induced* by χ^* if $\chi(n) = \chi^*(n)$ for all $(n, k) = 1$. A Dirichlet character modulo k is called *primitive* if it is not induced by a character of strictly smaller modulus k^* ($< k$).

2. Let $\varphi^*(k)$ denote the total number of primitive characters to modulus k . Prove that

$$\varphi^*(k) = \sum_{d|k} \mu(d) \varphi(k/d).$$

Furthermore, prove that the right hand side is equal to $k \prod_{p|k} (1 - \frac{2}{p}) \prod_{p^2|k} (1 - \frac{1}{p})^2$. (The notation $p \parallel k$ means that $p|k$ but $p^2 \nmid k$ for $\ell \geq 2$.)

3. Let $\chi \pmod{k}$ be a Dirichlet character, and let $k^*|k$.

- Show that χ is induced by a character $\pmod{k^*}$ if and only if $\chi(n_1) = \chi(n_2)$ whenever $(n_1, k) = (n_2, k) = 1$ and $n_1 \equiv n_2 \pmod{k^*}$.
- Show that χ is induced by a character $\pmod{k^*}$ if and only if $\chi(n_1) = 1$ whenever $(n_1, k) = 1$ and $n_1 \equiv 1 \pmod{k^*}$.

4. Let $k = 2^\alpha$. Suppose that $\alpha \geq 3$. Prove that

$$\chi(n; k, m_{-1}, m_0) = \begin{cases} 0, & \text{if } (n, 2) > 1, \\ (-1)^{m_{-1}r_{-1}} e\left(\frac{m_0 r_0}{2^{\alpha-2}}\right), & \text{if } (n, 2) = 1, \end{cases}$$

is a *primitive* character if and only if $(m_0, 2) = 1$.

5. Verify that the only real primitive character modulo 4 is given by

$$\chi_4(n) = \begin{cases} 0, & \text{if } (n, 2) > 1, \\ (-1)^{\frac{n-1}{2}}, & \text{if } (n, 2) = 1, \end{cases}$$

and the only real primitive characters modulo 8 are given by

$$\chi_8(n) = \begin{cases} 0, & \text{if } (n, 2) > 1, \\ (-1)^{\frac{n^2-1}{8}}, & \text{if } (n, 2) = 1, \end{cases}$$

and

$$\chi_4(n)\chi_8(n) = \begin{cases} 0, & \text{if } (n, 2) > 1, \\ (-1)^{\frac{(n-1)(n+5)}{8}}, & \text{if } (n, 2) = 1. \end{cases}$$

6. Let $f : \mathbb{Z}/q\mathbb{Z} \rightarrow \mathbb{C}$ be a complex valued function on the group $G = \mathbb{Z}/q\mathbb{Z}$.

Definition. Define the (discrete) Fourier transform of f , denoted by $\hat{f} : \mathbb{Z}/q\mathbb{Z} \rightarrow \mathbb{C}$, as

$$\hat{f}(\alpha) := \frac{1}{\sqrt{q}} \sum_{a \bmod q} f(a) e^{2\pi i \frac{\alpha a}{q}}.$$

- a) If f and g are two functions on the group G , show that

$$\sum_{a \bmod q} f(a) \overline{g(a)} = \sum_{\alpha \bmod q} \hat{f}(\alpha) \overline{\hat{g}(\alpha)}.$$

- b) Thus, show that

$$\sum_{a \bmod q} |f(a)|^2 = \sum_{\alpha \bmod q} |\hat{f}(\alpha)|^2.$$

- c) If $I = [A, B] \subset [0, q]$ is an interval, compute the Fourier transform $\hat{\mathbb{1}}(\alpha) : \mathbb{Z}/q\mathbb{Z} \rightarrow \mathbb{C}$ of the indicator function

$$\mathbb{1}_I(n) = \begin{cases} 1, & \text{if } n \in I \\ 0, & \text{otherwise} \end{cases}$$

of the interval I .

- d) Show that

$$\sum_{\alpha \bmod q} |\hat{\mathbb{1}}(\alpha)| \ll \sqrt{q} \log q.$$

- e) Applying the above results, prove the *Pólya-Vinogradov* bound, i.e., show that for any primitive Dirichlet character χ modulo q and any $N \geq 1$

$$\sum_{n=1}^N \chi(n) \ll \sqrt{q} \log q.$$

7. If p is a prime number, the quadratic residue symbol modulo p is defined by

$$\left(\frac{n}{p}\right) = \begin{cases} 0, & \text{if } n \equiv 0 \pmod{p}, \\ 1, & \text{if } n \equiv x^2 \pmod{p}, \text{ for some } x \\ -1, & \text{otherwise.} \end{cases}$$

- a) Show that the quadratic residue symbol modulo p is a multiplicative character on $(\mathbb{Z}/p\mathbb{Z})^\times$.
b) Use the *Pólya-Vinogradov* bound to show that the smallest quadratic non-residue (i.e., is not a square) modulo p is at most $O(\sqrt{p} \log p)$.

Exercise sheet 6

1. Let G be a finite abelian group and let $\widehat{G} = \{\chi : G \mapsto \mathbb{C}^\times \mid \chi \text{ character}\}$ be its *dual group* (the set of all characters of G). Prove that the two groups are isomorphic: $G \cong \widehat{\widehat{G}}$.

Hint. Think of $G = (\mathbb{Z}/q\mathbb{Z})^\times$.

2. The Riemann zeta function is given by

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s} \quad (\operatorname{Re}(s) > 1).$$

- (a) Use the Euler–Maclaurin formula (or Abel summation) to show that: for $\operatorname{Re}(s) > 1$

$$\zeta(s) = \frac{1}{s-1} + 1 - \int_1^{\infty} \frac{\{\xi\}}{\xi^{s+1}} d\xi$$

where $\{\xi\}$ denotes the fractional part of ξ .

This gives a meromorphic continuation of $\zeta(s)$ to the half-plane $\operatorname{Re}(s) > 0$.

- (b) Show that the Laurent series of $\zeta(s)$ about $s = 1$ is given by

$$\zeta(s) = \frac{1}{s-1} + \gamma + O(|s-1|),$$

where

$$\gamma = \lim_{n \rightarrow \infty} \left(\sum_{k=1}^n \frac{1}{k} - \log n \right)$$

is Euler's constant.

3. From Mertens' Theorem we see that

$$\prod_{p \leq x} \left(1 - \frac{1}{p}\right)^{-1} = C \log x + O(1)$$

for a certain constant $C > 0$. Show that

$$C = e^\gamma$$

where γ is Euler's constant.

4. Let $\chi \bmod q$ be a non-principal character. Let $x \geq 2$. Prove the following facts

a)

$$\sum_{n \leq x} \frac{\chi(n) \log n}{n} = -L'(1, \chi) + O_\chi \left(\frac{\log x}{x} \right).$$

Here O_χ means that the implied constant depends on χ (hence on q).

b)

$$\sum_{n \leq x} \frac{\chi(n) \log n}{n} = L(1, \chi) \sum_{m \leq x} \frac{\chi(m) \Lambda(m)}{m} + O_\chi \left(\frac{1}{x} \sum_{m \leq x} \Lambda(m) \right).$$

In particular, deduce from above (recall that $L(1, \chi) \neq 0$)

$$\sum_{m \leq x} \frac{\chi(m) \Lambda(m)}{m} = O_\chi(1). \quad (1)$$

c) From the upper bound (1), show that

$$\sum_{p \leq x} \frac{\chi(p) \log p}{p} = O_\chi(1).$$

d) Prove that

$$\sum_{p \leq x} \frac{\chi(p)}{p} = b(\chi) + O_\chi \left(\frac{1}{\log x} \right),$$

where

$$b(\chi) = \log L(1, \chi) - \sum_p \sum_{k \geq 2} \frac{\chi(p^k)}{kp^k}.$$

In particular, the sum $\sum_p \frac{\chi(p)}{p}$ converges.

5. Fix $\delta > 0$. Show that, for $|\arg(s)| \leq \pi - \delta$,

$$\frac{\Gamma'(s)}{\Gamma(s)} = \log s - \frac{1}{2s} + O_\delta \left(\frac{1}{|s|^2} \right).$$

Hint. Perhaps one needs to apply Cauchy's integral formula.

Exercise sheet 7

1. For any $\alpha > 0$, construct an entire function f_α of order exactly equal to α .

Hint. Let $f_\alpha(z) := \sum_{j=1}^{\infty} \frac{z^j}{\Gamma(1+j/\alpha)}$. Show that $f_\alpha(z)$ is of order α .

2. Are there entire functions of order 0 which are not polynomials?

3. Let $f(s)$ be an entire function of order α . For any given $\varepsilon > 0$, show that the sum

$$\sum_{\rho} \frac{1}{|\rho|^{\alpha+\varepsilon}}$$

is convergent. Here ρ runs over zeros of $f(s)$, counted with multiplicity.

4. By using logarithmic derivative of $\xi(s)$ and the functional equation $\xi(s) = \xi(1-s)$, show that the constant B in the Hadamard factorization $\xi(s) = e^{Bs} \prod_{\rho} \left(1 - \frac{s}{\rho}\right) e^{\frac{s}{\rho}}$ can be expressed as

$$B = - \sum_{\rho} \frac{1}{\rho} = -2 \sum_{\gamma > 0} \frac{\beta}{\beta^2 + \gamma^2},$$

Here

$$\sum_{\rho} \frac{1}{\rho} = \lim_{x \rightarrow \infty} \sum_{|\rho| \leq x} \frac{1}{\rho}.$$

5. Let χ be a primitive Dirichlet character mod q such that $\chi(-1) = 1$. Show that

$$L(s, \chi) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^s}$$

can be analytically continued to the whole complex plane and it satisfies the functional equation

$$\pi^{-\frac{1-s}{2}} q^{\frac{1-s}{2}} \Gamma\left(\frac{1-s}{2}\right) L(1-s, \bar{\chi}) = \frac{\sqrt{q}}{\tau(\chi)} \pi^{-\frac{s}{2}} q^{\frac{s}{2}} \Gamma\left(\frac{s}{2}\right) L(s, \chi).$$

What goes wrong when $\chi(-1) = -1$?

Hint. Let $\theta(x, \chi) = \sum_{n \in \mathbb{Z}} \chi(n) e^{-\pi n^2 \frac{x}{q}}$. Start with the integral representation

$$2\pi^{-\frac{s}{2}} q^{\frac{s}{2}} \Gamma\left(\frac{s}{2}\right) L(s, \chi) = \int_0^{\infty} \theta(x, \chi) x^{\frac{s}{2}} \frac{dx}{x},$$

and make use of the functional equation for $\theta(x, \chi)$:

$$\theta\left(\frac{1}{x}, \chi\right) = \frac{\sqrt{xq}}{\tau(\bar{\chi})} \theta(x, \bar{\chi})$$

that follows from the Poisson summation formula.

Exercise sheet 8

1. Assume that there exists a constant $C > 0$ such that any zero ρ of $\zeta(s)$ satisfies

$$\Re(\rho) < 1 - \frac{C}{\log(2 + |\Im(\rho)|)}.$$

Without using the explicit formula, show that there exists a constant $c > 0$ such that

$$\psi(x) = \sum_{n \leq x} \Lambda(n) = x + O\left(xe^{-c\sqrt{\log x}}\right).$$

Hint: Use Perron's formula along with Cauchy's residue theorem. You may shift the contour from $\sigma = 1 + \frac{1}{\log x}$ to $\sigma = 1 - \frac{C}{2(\log T)}$ and choose T optimally.

2. Let $\mu(n)$ be the Möbius function. Apply Perron's formula to show that

$$\sum_{n \leq x} \mu(n) = O\left(xe^{-c\sqrt{\log x}}\right),$$

where c is some positive constant.

Hint. You may use without giving a proof the following fact.

Fact. From the fact that $\zeta(\sigma + i\tau) \neq 0$ for $\sigma \geq 1 - c/\log(|\tau| + 2)$ one can show that

$$\frac{1}{|\zeta(\sigma + i\tau)|} \ll_c \log(|\tau| + 2) \quad \text{when} \quad \sigma \geq 1 - \frac{c/2}{\log(|\tau| + 2)}.$$

3. For any $N \in \mathbb{N}$, show that

$$\text{Li}(x) = \frac{x}{\log x} \sum_{n=0}^{N-1} \frac{n!}{(\log x)^n} + O_N\left(\frac{x}{(\log x)^{N+1}}\right).$$

Remark: This shows that, it is not true that $\pi(x) = \frac{x}{\log x} + O\left(\frac{x}{(\log x)^{2.1}}\right)$.

4. Show that

$$\sum_{n \leq x} \mu(n) = O(x^{1-\theta+\epsilon})$$

if and only if

$$\sum_{n \leq x} \Lambda(n) = x + O(x^{1-\theta+\epsilon}).$$